

Pause a simulation. Try every what-if from that moment.

DeepCausality CFD stops a running flow simulation at a moment you define, copies its exact state into branches, changes one decision in each, and runs them all at once. One table compares the outcomes.

THE PROBLEM

A run finishes and someone asks what would have happened with a different setting: three more degrees of bank, a hot-day atmosphere, ignition half a second later. The usual answer is another run, from the start or from a restart file saved at a time chosen in advance. Each answer repeats the shared past, and the runs share nothing.

WHAT IT DOES

A program **marches** a coupled simulation until a condition on the flow becomes true, **pauses** and receives the complete state, **forks** it into one branch per alternative, and **compares** the branches, which run at the same time, in one table with pass or fail gates. Flow, chemistry, navigation and control run in one process over one shared state, so a change in one stage reaches the others. A counterfactual is a what-if that starts from something that did happen.

The example. A vehicle re-enters at Mach 28 and loses GPS inside a plasma sheath. The run forks 17 branches from the moment the link drops, each with a different bank angle. The example places its aim point 30 m to the side of the unsteered path, so the sweep has a gap to close; the best branch closes it to 2.87 m. The whole example, from descent to reacquisition, takes 45.9 s on an Apple M3 Max, on a 32×32-cell flow layer.

MEASURED, ON AN APPLE M3 MAX

Whole powered descent	335.3 s , 5,425 coupled steps, four legs, sixteen gates
Peak memory	29.5 MB resident for the whole descent, including a five-branch fan-out
Corridor example	45.9 s for the descent, 17 branches and the legs after them, against a 600 s budget
Fan-out cost	1.16× one trunk step for five branches
Machine	Apple M3 Max, 16 cores, 128 GB

No cluster and no scheduler. Concurrent branches give results bit-identical to running them one after another; a regression test holds that.

CHECKED AGAINST REFERENCES

Manufactured solutions	observed order 1.98–2.01 against 2.00; divergence-free by construction
Taylor–Green (1937)	right-hand side matches to 1.11e-16
Sod shock tube	density error 0.0175 against a 0.03 tolerance
RAM-C II flight, NASA (1970)	peak electron density +0.22 decades from the flight value at 71.93 km, inside a ±0.70 allowance

Fourteen self-verifying targets, each exiting nonzero when its check fails. The references are exact solutions, published tables and flight data; no row compares against a run of Fluent, SU2 or OpenFOAM.

WHAT IT DOES NOT DO

- **It does not wrap your solver.** Branches run on this library's own solvers. It reads no mesh format; geometry is declared in code. It is Rust only, with no Python or C interface.
- **No turbulence models yet.** Validated incompressible cases run at Re 100 to 1600. The what-if examples run on 32×32 cells, and nothing has run at production mesh scale.
- **The retropropulsion drag comes from a cited correlation** (Jarvinen and Adams, 1970). The simulated plume did not reproduce the drag collapse on this solver; the measurement is published.
- **One machine.** Distributed execution is a stated non-goal, and GPU work is deferred.

GETTING STARTED

One command runs the corridor example from a clean clone, in 45.9 s after a first-time build. It needs Rust 1.98 or newer; skip the first line if you already have it.

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
git clone --depth 1 https://github.com/deepcausality-rs/deep_causality.git
cd deep_causality
cargo run --release -p avionics_examples --example plasma_blackout_corridor
```

Keep `--depth 1`: the full history is 2.46 GB against 195 MB shallow. As a library, cargo add `deep_causality_cfd`. MIT licensed. Part of the DeepCausality project, an LF AI & Data Foundation Sandbox project maintained by the Center for Dynamic Causality.